

Thilakkumar V

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EDUCATION

National Institute of Technology, Tiruchirappalli <i>Bachelor of Technology in Civil Engineering, Minors in Computer Science</i> <i>Relevant Coursework: Data Structures, DBMS, Artificial Intelligence, Operating Systems, Software Engineering</i>	2020 – 2024 8.21 CGPA
Shree Niketan Matric. Hr Secondary School, Chennai <i>Class XII, Central Board of Secondary Education</i>	2020 91.17%

EXPERIENCE

Product Engineer (Backend & DevOps) Ganit Inc. <i>An AI-first decision science company delivering enterprise-grade data solutions across industries</i>	Aug 2024 – Present Chennai, TN
- Engineered scalable backend systems and real-time APIs using FastAPI, Django, and WebSockets for high-throughput applications in FMCG, HR tech, and healthcare. - Built 7 real-time Sales Tower Dashboards for an FMCG giant using Node.js and Google BigQuery; defined 20+ KPIs, authored 50+ optimized SQL queries, and improved performance by 40% via code refactoring and caching. Ensured robustness with Mocha, Chai, and NYC. - Managed CI/CD workflows and DevOps pipelines for 4 production healthcare apps (Django + FastAPI) using GitHub Actions, Docker, and AWS, achieving 2x task throughput and reducing deployment errors by 80%. - Developed an AI-powered Resume Parser Tool using Gemini API, OneDrive Graph APIs, and LLMs, built with Django REST; deployed via AWS EC2 & Amplify, achieving 90% accuracy and 75% manual effort reduction. - Automated cloud infrastructure using Terraform and AWS CloudFormation; provisioned EC2, RDS, S3, VPC, reducing infrastructure deployment time by 60% and minimizing human error. - Built real-time backend for DatForge AI Chatbot using FastAPI & WebSockets, achieving sub-200ms latency, >99.9% uptime, and support for 1000+ concurrent users; integrated with AWS Bedrock GENAI.	
Technical Product Analyst Intern BlockTrack <i>An IIT Madras incubated Health-tech startup specializing in Blockchain and SaaS products</i>	May 2023 – Jul 2023 Chennai, TN
- Conducted data-driven analysis on 50+ market players, identifying trends and enabling tech partnerships. - Leveraged PEST analysis to refine market positioning, boosting profit margins by 5% and market share by 15%. - Designed and prototyped modular insurance workflows using BPMN tools and backend logic schemas, projected to enable a 75% expansion in user adoption. - Analyzed 2000+ structured survey responses using Python (NumPy, Seaborn), generating actionable insights that informed 6 high-impact strategic initiatives.	

PROJECTS

NexEvent – Event Management Platform React.js, Tailwind CSS, Django REST, Docker, AWS	Jun 2024 – Jul 2024
- Designed and deployed a full-stack event management platform supporting 5+ admin features and 10K+ concurrent users, enabling efficient event creation, guest tracking, and real-time attendance. - Architected and implemented RESTful APIs with Django REST Framework to handle CRUD operations, RSVP logic, reminders, and role-based access control. - Built a responsive and dynamic frontend using React.js and Tailwind CSS, reducing page load time by 40% and enhancing user experience across devices. - Automated the end-to-end CI/CD pipeline using Docker, GitHub Actions, and AWS CodePipeline; deployed to ECS Fargate with autoscaling, achieving 99.9% uptime and 0-downtime releases. - Integrated CloudWatch for real-time monitoring and alerting, reducing debugging time by 70% and enabling proactive issue resolution.	
Predictive Drought Risk Model Python, ArcGIS, MLR, SVM, Data Studio	Jan 2024 – May 2024
- Integrated historical data & EC-Earth3 projections under CMIP6 framework, covering 6500+ rain gauge stations. - Enhanced predictive accuracy by 20% with advanced bias correction techniques (MLR, SVM) - Automated 7.5 lakh SPI iterations to quantify drought characteristics in the Cauvery River Basin. - Designed interactive dashboards and spatial visualizations using Google Earth Engine for actionable insights.	

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, Bash, SQL (MySQL, PostgreSQL)

Frameworks: Django, FastAPI, ReactJS, Node.js, Express.js, Tailwind, Bootstrap

Cloud/DevOps: AWS (EC2, S3, RDS, CloudWatch, Amplify, Bedrock), GCP, Docker, GitHub Actions, Terraform, Jenkins

Tools & Platforms: Git, Postman, VS Code, Linux, WebSockets, REST APIs, GraphQL, CI/CD, Microservices, CRON Jobs

Certifications: AWS Certified Cloud Practitioner

LEADERSHIP / ACTIVITIES

Chairperson, Nittfest'23 | *The Annual Cultural Festival of NIT Trichy*

Mar 2022 – May 2024

- Led and coordinated a **3-tier team** of **250+ members**, executing 80+ events for 12000+ students.
- Managed a **75 lakh INR** budget with strategic marketing, promotions, and risk management.
- Organized events featuring high-profile guests like **Dr. Kamal Haasan**, partnering with **7+ production houses**.
- Increased corporate **sponsorship by 70%** through innovative branding and outreach for **50+ companies**.

Consultant | *180 Degrees Consulting, NIT Trichy*

Aug 2022 – May 2024

- Handled 4 consulting projects (3 startups, 1 NGO), with an **NPS of 100%** across clients.
- Within the **FinTech Innovation Project - BharatX**, conducted primary research on 10+ professional education loan systems, analyzed 5-year Indian Ed-Tech market growth with 7 parameters, and **devised 3 innovative strategies**.

ACHIEVEMENTS

- **Secured 2nd Position** at TransfiNITTe'23, a 42-hour long hackathon on software projects

Oct 2023

- **Winner** - Bid and Build, eSummit'23, IIT Bombay

Feb 2023